INCH-POUND

MIL-P-70506A (AR)
AMENDMENT 1
3 April 1989

MILITARY SPECIFICATION

PROJECTILE, 155MM: XM864 BASE BURNER ASSEMBLY
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-70506A (AR) dated 4 April 1988, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Add the designation "(M864)" in the title and change XM864 to "M864" wherever it appears in the specification.

PAGE 2

2.1.1 Add the following military specification:

"MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification"

Add the following military standard:

"MIL-STD-331A - Test 105.1, Temperature and Humidity"

PAGE 3

Add the following new paragraphs as follows:

"2.2 Non-Government publications. The following document (s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise speciried, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitaion (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z1.1 - Guide for Quality Control Charts
ANSI Z1.2 - Control Chart Method of Analyzing Data
ANSI Z1.3 - Control Chart Method of Controlling Quality During Production

(Applications for copies should be addressed to American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3.3.1 Change the paragraph to read as follows:

"3.3.1 Loose igniter composition. After consolidation there shall be no visual evidence of loose igniter composition."

PAGE 5

3.8 Delete the second sentence and substitute the following:

"They shall be free of burrs, sharp edges, cracks, gouges, cuts, dirt, grease, rust, and other foreign matter, unless defined elsewhere in this specification."

3.8.1 Change title to read:

"3.8.1 Tracer cups."

3.8.2 Change the paragraph to read:

"3.8.2 Igniter cup, loading assembly. The stepped area of the finished igniter cup need not be perfect form. Damaged areas up to 25 percent of the total surface area are allowed. Damage in excess of 25 percent is cause for rejection."

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Add the following new paragraphs:

"3.9 Weather seal assembly. The weather seal assembly shall not allow moisture to leak into the base burner assembly.

3.9.1 Weather seal resistance to air pressure. The weather seal assembly shall withstand an air pressure test of 15 psi for 15 seconds.

3.9.2 Weather seal shelf life in storage. The shelf life for production weather seals in storage shall be a minimum of one year from the date of their production.

3.9.3 Weather seal degradation. Prior to their use, if it is detected by the base burner manufacturer that a shipment of weather seal assemblies or parts thereof has been contaminated due to exposure to a foreign substance or by the degradation of the packaging assembly, those weather seal assemblies will be rejected and will not be permitted for use.

NOTE: A foreign substance is considered to be any unknown substance; i.e., liquid, powder, gel, residue, etc. that does not conform to the provisions of the weather seal assembly drawing, or visually compare in quality, or physical condition to prior shipments.

3.9.4 Weather seal adhesion. The weather seal will not leak when tested in accordance with this specification."

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4.3.1.1 Examination or Test, delete:

"Ingiter
(Dwg. 12630572)" and substitute:

"Igniter Cup, Loading Assembly
(Dwg. 12630572)"

PAGE 8

TABLE I. First Article Inspection - Weather Seals, Add:

"Weather Seal Adhesion Test 6 3.9.4 4.5.10"

PAGE 9

4.4.1 Add the following requirement:

"f. Weather seal assemblies from one lot from one manufacturer."

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Add the following new paragraph:

"4.4.1.1 Weather seal assembly raw materials. The raw materials used in the construction of the M864 weather seal assembly will be subject to the following requirements:

a. Aluminum from one lot from one manufacturer.

b. Polyvinyl fluoride (disk seal) from one lot from one manufacturer.

c. Adhesive from one lot from one manufacturer.

NOTE: Each lot of weather seal assemblies shall be constructed from a batch of adhesive, a lot of aluminum, and a lot of polyvinyl fluoride dedicated solely for the production of that one lot of weather seals. All remaining amounts of raw materials, following the production of that lot of weather seals shall be discarded."

PAGE 9

4.4.2 Add the following new paragraph:

"4.4.2 c. Statistical process control (SPC). For those dimensions identified in the Classification of Defects Tables by the reference "(see Appendix A)", the contractor must utilize the principals of SPC as defined herein, ANSI Z1.1, ANSI Z1.2, ANSI Z1.3, and in Appendix A. Then the contractor satisfies the control conditions identified in Appendix A, acceptance will then be by SPC. The SPC sample size and frequency of selection shall be determined by the contractor. If the SPC sample is to be less than 4 items per hour, then the approval of the procuring agency must be sought prior to the elimination of sampling inspection for acceptance. The contractor may request to implement SPC acceptance for any other feature that is not annotated in the classification for defects by requesting approval of the procurement agency and by meeting all the conditions specified in this specification. When not accepting product by SPC, acceptance of product for those features identified with references to Appendix A, will only be in accordance with MIL-STD-414 or MIL-STD-1235 plans CSP-1, -2, -T using variable measurement capable gages."

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4.4.2.2 Change critical defect to read:

"Critical Evidence of loose igniter compositon 100% 3.3.1 Visual/Note 2"

4.4.2.2 Examination or test. Add "(see Appendix A)" after major defects 105, 106 and 107.

Change Note 1 to read:

"1. Damaged or missing cadmium plating is permissable on tracer cup provided that the base metal is not gouged or cracked. Flakes of cadmium plating on surface of igniter material is permissible."

Add to the bottom of paragraph 4.4.2.2 under Notes:

"2. As a referee test for possible loose igniter composition, refer to paragraph 6.9."

PAGE 12

4.4.2.3 Make the following changes:

Delete Major 101 in its entirety.

Delete Major 102 and substitute the following:

"Major 101 Plug missing 100% 3.1 Visual"

Add:

"Special
a Igniter cup missing or incorrectly oriented (Note 1) 100% 3.1 Visual"

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4.4.2.4 Delete Major 102 in its entirety. Add the following:

"Special
a Igniter cup missing or incorrectly oriented (Note 1) 100% 3.1 Visual"

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PAGE 16

4.4.2.7 Add the following special defect:

"Special					
a	Weather seal adhesion test	6	3.9.4	4.5.10"	

PAGE 18

4.4.2.9 Delete minors 201, 202, 203 in their entirety and substitute the following:

"Minor				
201	"M864 BASE BURNER ASSEMBLY" notation and lot number missing, misleading or unidentifiable	.65%	3.1	Visual
202	Propelling grain batch number missing, or unidentifiable	.65%	3.1	Visual
203	Evidence of poor workmanship	.65%	3.8	Visual"

PAGE 24

Add the following new paragraphs:

"4.5.10 Weather seal adhesion test. An adhesion test shall be conducted on each lot of weather seal assemblies. From each lot, a sample of weather seals shall be chosen at random and assembled onto six M864 base assembly test units. On each of the test units, one (1) weather seal shall be applied to the outside opening of each base closure.

4.5.10.1 Assembly of test units.

a. The test units will be assembled without propellant grains, igniter cups of M864 base burner metal parts internal hardware. The test units will be assembled with their two o-rings present, minus the internal weather seal, and the assembly will be torqued shut to M864 base burner assembly loading requirements.

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b. Weather seals picked randomly from the lot, will be applied to the outside of each of the base closures. Each weather seal will be applied over the outside opening of the base closure using the normal application process. The base closures will then be serialized in two inch block letters with an indellible ink pen, around the outside circumference of each base closure to help distinguish the different test units after the temperature-humidity test is completed.

c. A pre-weighed 100.00 + 5.00 gram amount of desiccant (MIL-D-3464, Type II), enclosed in a porous container bag, shall then be added to the empty internal propellant grain area of each test unit base. Immediately prior to loading, each desiccant package shall be baked dry, and then weighed and recorded. Each bag of desiccant will be assigned to a numbered test unit. The weight of each bag of desiccant shall be recorded with the number of the test unit it is placed in. The entire weighing and loading process for the test units shall be conducted in a low-humidity, climate controlled room to prevent the absorption of moisture. A temperature-humidity resistant sealant (LOC-TITE brand, SUPER FLEX ULTRA-BLUE, vendor no: #587 or equivalent) may be applied to the base-base closure gap of each of the six units during the loading process to prevent any possible moisture leakage into the unit through this channel during the test.

4.5.10.2 Weather seal assembly air pressure test. Prior to the start of the adhesion test, each test unit will be subjected to the following examination. An air pressure test in accordance with 4.5.6 shall be conducted where the pressure inside the air pressure test chamber must achieve 15 psi and maintain that level for 15 seconds. Failure to meet either criteria shall be cause for rejection of the test quantity. In the event that a test unit should fail, an investigation into the cause of the failure shall be initiated and a course of corrective action shall be reported to the cognizant engineering facility for approval before retest can occur.

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4.5.10.3 Orientation of test units. The assemblies shall be placed on a support during the testing. The support shall be configured to allow air to circulate above, around, and below the support and test units. The assemblies shall be placed in an upright position on the support, with the weather seal on the bottom, facing the support. The support shall not be allowed to come in direct contact with the weather seals themselves at any time. Careful consideration should be used if there is a need to move the test units during the test, so that their orientation inside the test environment is not drastically changed.

4.5.10.4 Test procedure. The test shall be conducted in accordance with MIL-STD-331, Test 105.1, paragraph 5.1. through 5.1.18, Two Chamber Method, or as an option, 5.2 through 5.2.3, Single Chamber - Cycling Method. After successful completion of the air pressure test, the six desiccant filled test units will be placed in an environmental weathering cabinet, and cycled through the variated 28 day temperature and humidity schedule.

NOTE: It is required that the pH of a condensed sample of vapor from the test chamber be between 6.0 and 7.3 at 63 degrees F. It is recommended that the specified humidity for the weather seal adhesion test be obtained through the use of steam or by vaporizing distilled, demineralized or deionized water.

4.5.10.5 Post - test examination. After the test items have been conditioned by the above procedure, they shall undergo the following examinations. The test units shall be disassembled and visually examined for corrosion, moisture and other signs of deterioration along the inside chamber of the test unit. The desiccant package shall be carefully removed from the internal chamber and weighed for percent moisture content on a scale accurate to within 0.001 (one one-thousandth) grams. All desiccant weight increases, if any, will be recorded along with the test unit number that that particular desiccant package was taken from. Any weight increase in any of the desiccant packages which is greater than one percent (1%) of the initial weight of that particular package from the start of the test, shall be cause for rejection of that weather seal lot and must be reported to the cognizant engineering facility."

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Add the following new paragraphs:

"5.3 Storage of approved weather seal lots. Upon completion of a pre-determined quantity of production weather seal assemblies, the seal assemblies shall be packed in a government approved storage container to prevent decontamination of the seals. As the seals come off of the production line, they will have a maximum allowable time of six (6) hours dormancy before being required to be packed for storage.

5.3.1 Date of manufacture. The date of manufacture, the weather seal lot number, and the name of the manufacturer for that weather seal lot will be recorded and included with each container of weather seals."

PAGE 25

Add the following new paragraphs:

"6.9 Drop test for determining loose igniter composition. If visual inspection is questionable for determining loose igniter composition the following test will be implemented:

Place the subject igniter cup face down (open end down) onto a clean, flat piece of blank, unlined white paper. Manually raise the subject igniter cup face down above the paper and release it, allowing it to free fall onto the paper. The igniter cup should be dropped from a height of one-half inch (.5 inch) approximated by eye, so that the igniter cup strikes the paper face down. If any igniter composition is deposited on the paper, that igniter cup will be rejected. At the start of each new test, a new piece of paper will be used."

PAGE 26

Add new Appendix A included herein, pages (26) through (28).

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Preparing activity:
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(Project 1320-AB61)

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APPENDIX A

10. SCOPE

10.1 Scope. This appendix details the procedure for implementation of statistical process control for articles covered in this specification. This Appendix is a mondatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Application of statistical process control. The contractor will for the identified dimensions maintain Xbar and R SPC charts. After the contractor has demonstrated that his process for manufacturing a particular feature is in control, as defined in paragraph A.1.1; he will discontinue acceptance sampling inspection in favor of acceptance by SPC. However, SPC charting will be maintained for all identified dimensions whether or not the process is in control. It is expected that the contractor will make every effort to bring each process into control. It is required that the contractor re-evaluate his established control limits on a monthly basis and make those changes warranted by the data. At no time should his process parameters vary such that it will fail to meet the requirements established in subsequent paragraphs. Should this happen, the contractor will return to acceptance by sampling.

30.1.1 Implementation of SPC for acceptance of product. For those dimensions identified in the classification of defects tables as being applicable for elimination of lot sampling acceptance, the contractor shall implement, on a dimension by dimension basis, acceptance by SPC after he conforms to each of the following steps:

a. The contractor performs an indepth process capability study to establish the mean process average (Xdouble bar), Rbar, upper and lower control limits. This study will consist of at least 40 subgroups of data collection.

b. The contractor demonstrates that his process is under control for at least 20 consecutive production days. Control is defined as remaining within the established control parameters and not exhibiting any of the out of control conditions defined in paragraph A.1.2.a through A.1.2.e more than two times on an individual character basis.

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c. Thirty days consecutive production have passed acceptance sampling for the feature.

d. The process has a demonstrated Cpk equal to or greater than 1.3 as demonstrated when conducting the above test. Cpk is defined as follows:

$$Cpk = \frac{\text{process mean - nearest specification limit}}{3 \text{ sigma}}$$

sigma = process standard deviation (calculated by the range method per the referenced ANSI documents)

e. The contractor documents his detailed process description. This process description must be submitted to Commander, AMCCOM, Attn: AMSMC-QAR-Q (D), Picatinny Arsenal, New Jersey 07806-5000 for review and disapproval if the description does not meet its objectives. That is the process description will identify all known contributing variables and their respective controls that are necessary to maintain the specific dimension (i.e., weight, temperature, machine speed, time, tool life, solution proportion, etc.). The government will furnish the contractor with written notice of the acceptability of his process controls. A notice of control acceptability is required prior to the implementation of acceptance by SPC.

30.1.2 Process control analysis. Once SPC acceptance is implemented, the contractor will evaluate his process for evidence of nonrandom variation (e.g. trends, extreme values, etc.). Should any out of control condition be detected, the product produced from the last sample till the process is brought back into control will be isolated for analysis. The contractor will bring the process back into control or shut the process down. Upon determining the cause of the out of control condition, the contractor will determine if the product manufactured and isolated could be out of engineering tolerance. If the out of control condition did not result in the production of nonconforming material, the material will be released. If the out of control condition could have created nonconforming product, the contractor will sample the product using MIL-STD-105 and the AQL specified in the classification of defects table to determine the acceptability of the isolated product. This evaluation of out of control product requires the review and approval of the local government quality assurance representative. As a minimum, the contractor will look for any of the following out of control indications:

a. If any point goes outside the Xbar or R chart control limits.

b. If 9 points in a row are either continually above or continually below the X double bar.

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c. If 6 points in a row are either continually increasing or continually decreasing on the Xbar chart.

d. If 6 points in a row are increasing on the range chart.

e. If any individual sample, Xi, falls outside of the specification limits.

30.1.2.1 Process capability review. At least once a month the contractor will review his process data and establish updated control limits.

30.1.3 Mandatory return to sampling acceptance. Acceptance by SPC will be discontinued in accordance with the following paragraphs. In any case, control charting will be continued. When this occurs, the contractor may reintroduce acceptance by SPC by repeating the steps outlined in paragraph A.1.1.

30.1.3.1 Out of control incidents. If for any individual process, an out of control incident is identified 3 times within 5 consecutive shifts or 5 times within 30 consecutive shifts; and the analysis indicated that nonconforming product may have been manufactured, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.3.2 Process interruption. If the process is interrupted for more than 10 calendar days, the contractor will return to acceptance by sampling for that pertinent feature.

30.1.3.3 Loss of control. If either by testing or data review it is determined that the Cpk is no longer at the required minimum of 1.3, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.4 Records. The contractor will maintain all records dealing with SPC in his files for a minimum of 5 years after completion of the contract.

30.2 Government auditing. In order to verify the state of control of the process, it is recommended that government representatives select samples at unscheduled times. The sample mean and range should then be compared to the data being recorded by the contractor. If the data is not consistent with the reported data, corrective action should be sought and if necessary, screening of material should be requried.

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